

Web programming

Laboratórios de Sistemas e Serviços

Mário Antunes

2026

Exercises

This guide accompanies the theoretical slides on Dynamic Web Pages. You will build a complete web application from scratch, starting with a static profile and evolving it into a dynamic system with user authentication, real-time maps, and chat support.

Technologies used:

- **Frontend:** HTML5, CSS3 (Nord Light Theme), Vanilla JavaScript.
- **Backend 1:** Node.js (Express) for Authentication and Chat.
- **Backend 2:** Python (FastAPI) for Geolocation and Data processing.
- **Infrastructure:** Docker & Docker Compose.

Phase 1: Project Setup & Static Structure

Step 0: Installation & Verification

Before writing code, ensure your environment is ready.

1. **Open your terminal.**
2. **Verify Docker:** `docker --version` and `docker compose version`
3. **Verify Node.js (Optional):** `node -v`

Step 1: Folder Structure

1. Create a main folder named `ex05`.
2. Inside it, create three subfolders: `frontend`, `auth-service`, and `geo-service`.
3. Create a `compose.yml` file in the root.

Step 2: The Base Docker Compose

Open `compose.yml` and paste this code:

```
services:
  # 1. Frontend Server (Nginx)
  web:
    image: nginx:alpine
    container_name: frontend_server
    ports:
      - "8080:80"
    volumes:
      - ./frontend:/usr/share/nginx/html
```

Step 3: The Static Profile (HTML)

Open `frontend/index.html`. Note the `chatWidget` div at the bottom; this will be used in Step 11.

```
<!DOCTYPE html>
<html lang="en">
```

```

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Dynamic Profile</title>
  <!-- Leaflet CSS for the Map (Step 15) -->
  <link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css" />
  <link rel="stylesheet" href="style.css">
  <script src="app.js" defer></script>
</head>
<body>
  <header>
    <h1>User Profile</h1>
    <nav>
      <button id="loginBtn">Login</button>
      <button id="logoutBtn" style="display:none;">Logout</button>
    </nav>
  </header>

  <main id="app">
    <!-- Login Modal -->
    <dialog id="loginDialog">
      <form id="loginForm">
        <h2>Welcome Back</h2>
        <input type="text" id="username" placeholder="Username" required>
        <input type="password" id="password" placeholder="Password" required>
        <button type="submit">Sign In</button>
        <button type="button" id="cancelLogin">Cancel</button>
      </form>
    </dialog>

    <!-- Content Section (Hidden until logged in) -->
    <div id="contentArea" class="hidden">
      <section class="card profile-card">
        
        <h2 id="welcomeMsg">Hello, User</h2>
        <p>Full Stack Student</p>
      </section>

      <section class="card gallery-card">
        <h3>Photo Gallery</h3>
        <div id="galleryGrid" class="grid"></div>
      </section>

      <section class="card map-card">
        <h3>Live Location Tracker (WebSocket)</h3>
        <div id="map"></div>
      </section>

      <!-- Chat Widget (For Step 11) -->
      <div id="chatWidget">
        <div id="chatHeader">Support Chat</div>
        <div id="chatMessages"></div>
        <input type="text" id="chatInput" placeholder="Type a message...">
      </div>

      <div id="guestMessage">
        <p>Please log in to view the dashboard.</p>
      </div>
    </main>

```

```

<footer>
  <p>&copy; 2025 Web Engineering</p>
</footer>

<!-- Leaflet JS -->
<script src="https://unpkg.com/leaflet@1.9.4/dist/leaflet.js"></script>
</body>
</html>

```

Step 4: Nordic Styling (CSS)

Open frontend/style.css. This defines the layout and the Chat Window position.

```

:root {
  --polar-night: #2E3440;
  --snow-storm: #ECEFF4;
  --frost-1: #8FBCBB;
  --frost-2: #88C0D0;
  --frost-3: #81A1C1;
  --frost-4: #5E81AC;
  --aurora-red: #BF616A;
}
body {
  font-family: 'Noto Sans', sans-serif;
  background-color: var(--snow-storm);
  color: var(--polar-night);
  margin: 0;
  display: flex;
  flex-direction: column;
  min-height: 100vh;
}
header {
  background-color: var(--frost-4);
  color: white;
  padding: 1rem 2rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
}
button {
  background-color: var(--frost-3);
  color: white;
  border: none;
  padding: 0.5rem 1rem;
  border-radius: 4px;
  cursor: pointer;
  font-weight: bold;
}
button:hover { background-color: var(--frost-2); }
main {
  flex: 1;
  padding: 2rem;
  max-width: 1200px;
  margin: 0 auto;
  width: 100%;
}
.card {
  background: white;
  padding: 1.5rem;
  border-radius: 8px;
  margin-bottom: 2rem;
}

```

```

    box-shadow: 0 2px 4px rgba(0,0,0,0.05);
}
.hidden { display: none; }
dialog {
    border: 1px solid var(--frost-2);
    border-radius: 8px;
    padding: 2rem;
}
dialog::backdrop { background: rgba(46, 52, 64, 0.5); }

/* Chat Widget Styling */
#chatWidget {
    position: fixed;
    bottom: 20px;
    right: 20px;
    width: 300px;
    background: white;
    border: 1px solid var(--frost-3);
    border-radius: 8px;
    overflow: hidden;
    display: flex;
    flex-direction: column;
    box-shadow: 0 4px 6px rgba(0,0,0,0.1);
}
#chatHeader {
    background: var(--frost-4);
    color: white;
    padding: 10px;
    font-weight: bold;
}
#chatMessages {
    height: 200px;
    padding: 10px;
    overflow-y: auto;
    font-size: 0.9rem;
    background-color: #fff;
}
#chatInput {
    border: none;
    border-top: 1px solid #eee;
    padding: 10px;
    outline: none;
}

/* Map & Gallery */
#map { height: 300px; width: 100%; border-radius: 4px; }
.grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(150px, 1fr));
    gap: 1rem;
}
.grid img { width: 100%; border-radius: 4px; }

```

Step 5: Basic JavaScript

Create frontend/app.js.

```

console.log("Application Loaded");

const loginBtn = document.getElementById('loginBtn');
const loginDialog = document.getElementById('loginDialog');

```

```
const cancelLogin = document.getElementById('cancelLogin');
```

```
loginBtn.addEventListener('click', () => loginDialog.showModal());  
cancelLogin.addEventListener('click', () => loginDialog.close());
```

Test Phase 1: Run `docker compose up -d` and visit `http://localhost:8080`.

Phase 2: Node.js Backend (Auth & Chat)

Step 6: Setup Node Service

1. Go to `auth-service/`.
2. Create `package.json`:

```
{  
  "name": "auth-service",  
  "main": "server.js",  
  "scripts": { "start": "node server.js" },  
  "dependencies": { "express": "^4.18.2", "cors": "^2.8.5", "ws": "^8.13.0" }  
}
```

3. Create Dockerfile:

```
FROM node:18-alpine  
WORKDIR /app  
COPY package.json .  
RUN npm install  
COPY . .  
EXPOSE 3000  
CMD ["npm", "start"]
```

Step 7: Implement Server Logic

Create `auth-service/server.js`.

```
const express = require('express');  
const cors = require('cors');  
const http = require('http');  
const WebSocket = require('ws');
```

```
const app = express();  
const server = http.createServer(app);  
const wss = new WebSocket.Server({ server });
```

```
app.use(cors());  
app.use(express.json());
```

// Login Endpoint

```
const VALID_USER = { username: "admin", password: "123" };  
app.post('/login', (req, res) => {  
  const { username, password } = req.body;  
  if (username === VALID_USER.username && password === VALID_USER.password) {  
    res.json({ success: true, token: "jwt-123" });  
  } else {  
    res.status(401).json({ success: false, message: "Invalid credentials" });  
  }  
});
```

// Chat WebSocket Logic

```
wss.on('connection', (ws) => {  
  ws.send('Support: Hello! How can I help you?');  
  ws.on('message', (message) => {  
    setTimeout(() => {
```

```

        ws.send(`Support: I received "${message}"`);
    }, 1000);
});
});

server.listen(3000, () => console.log('Auth/Chat running on 3000'));

```

Step 8: Update Frontend JS

Modify frontend/app.js to handle login.

```

// Add these references
const loginForm = document.getElementById('loginForm');
const contentArea = document.getElementById('contentArea');
const guestMessage = document.getElementById('guestMessage');
const logoutBtn = document.getElementById('logoutBtn');

loginForm.addEventListener('submit', async (e) => {
    e.preventDefault();
    const username = document.getElementById('username').value;
    const password = document.getElementById('password').value;

    try {
        const response = await fetch('http://localhost:3000/login', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ username, password })
        });
        const data = await response.json();

        if (data.success) {
            loginDialog.close();
            handleLoginState(true);
        } else {
            alert(data.message);
        }
    } catch (err) { console.error(err); }
});

function handleLoginState(isLoggedIn) {
    if (isLoggedIn) {
        contentArea.classList.remove('hidden');
        guestMessage.classList.add('hidden');
        loginBtn.style.display = 'none';
        logoutBtn.style.display = 'inline-block';

        // Load dynamic features
        loadGallery();
        initChat();
        initMap();
    } else {
        location.reload();
    }
}

logoutBtn.addEventListener('click', () => handleLoginState(false));

```

Step 9: Update Docker Compose

Update docker-compose.yml to include auth.

```

services:
  web:

```

```
# ... (existing config)
auth:
  build: ./auth-service
  container_name: auth_server
  ports:
    - "3000:3000"
```

Phase 3: Features (Gallery & Chat)

Step 10: The Photo Gallery (JS)

Append this to frontend/app.js:

```
function loadGallery() {
  const galleryGrid = document.getElementById('galleryGrid');
  const images = [
    'https://picsum.photos/id/101/300/200',
    'https://picsum.photos/id/102/300/200',
    'https://picsum.photos/id/103/300/200',
    'https://picsum.photos/id/104/300/200'
  ];
  galleryGrid.innerHTML = '';
  images.forEach(url => {
    const img = document.createElement('img');
    img.src = url;
    galleryGrid.appendChild(img);
  });
}
```

Step 11: The Chat Client (WebSocket)

Append this to frontend/app.js. This code makes the Chat Window defined in HTML/CSS functional.

```
function initChat() {
  const chatInput = document.getElementById('chatInput');
  const chatMessages = document.getElementById('chatMessages');

  // Connect to Node.js WebSocket
  const socket = new WebSocket('ws://localhost:3000');

  socket.addEventListener('message', (event) => {
    addMessage(event.data, 'server');
  });

  chatInput.addEventListener('keypress', (e) => {
    if (e.key === 'Enter') {
      const text = chatInput.value;
      socket.send(text);
      addMessage("You: " + text, 'user');
      chatInput.value = '';
    }
  });

  function addMessage(text, sender) {
    const div = document.createElement('div');
    div.innerText = text;
    div.style.textAlign = sender === 'user' ? 'right' : 'left';
    div.style.color = sender === 'user' ? '#5E81AC' : '#BF616A';
    div.style.marginBottom = '5px';
    chatMessages.appendChild(div);
    chatMessages.scrollTop = chatMessages.scrollHeight;
  }
}
```

```
}  
}
```

Phase 4: FastAPI Backend (Geolocation)

Step 12: Setup Python Service

1. Go to geo-service/.

2. Create requirements.txt:

```
fastapi  
uvicorn  
websockets
```

3. Create Dockerfile:

```
FROM python:3.9-slim  
WORKDIR /app  
COPY requirements.txt .  
RUN pip install --no-cache-dir -r requirements.txt  
COPY . .  
CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

Step 13: Implement Python Logic

Create geo-service/main.py.

```
from fastapi import FastAPI, WebSocket  
from fastapi.middleware.cors import CORSMiddleware  
import asyncio, random  
  
app = FastAPI()  
app.add_middleware(CORSMiddleware, allow_origins=["*"], allow_methods=["*"])  
  
@app.websocket("/ws/location")  
async def websocket_endpoint(websocket: WebSocket):  
    await websocket.accept()  
    lat, lon = 40.64427, -8.64554  
    try:  
        while True:  
            lat += random.uniform(-0.001, 0.001)  
            lon += random.uniform(-0.001, 0.001)  
            await websocket.send_json({"lat": lat, "lng": lon})  
            await asyncio.sleep(2)  
    except: print("Disconnected")
```

Step 14: Update Docker Compose

Add the geo service to docker-compose.yml.

```
services:  
  # ... existing web and auth ...  
  geo:  
    build: ./geo-service  
    container_name: geo_server  
    ports:  
      - "8000:8000"
```

Step 15: The Map (Leaflet + WebSocket)

Append this to frontend/app.js:


```

function initMap() {
  const map = L.map('map').setView([40.64427, -8.64554], 13);
  L.tileLayer('https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png', {
    attribution: '© OpenStreetMap contributors'
  }).addTo(map);

  const marker = L.marker([40.64427, -8.64554]).addTo(map);

  // Connect to Python WebSocket
  const geoSocket = new WebSocket('ws://localhost:8000/ws/location');

  geoSocket.addEventListener('message', (event) => {
    const data = JSON.parse(event.data);
    const newLatLng = [data.lat, data.lng];
    marker.setLatLng(newLatLng);
    map.panTo(newLatLng);
  });
}

```

Final Step: Run `docker compose up -d --build` and enjoy your app!

Phase 5 (Optional)

Implement two major changes in the code:

1. For a more realistic approach, the server should compare password hashes instead of the passwords directly. Implement this modification on the webpage and the authentication server.
2. Create multiple chat windows, updating the webpage and server to support this.